

Requirements Document

Project Name: Event Management System

Functional Requirements

1. User Authentication

- Users must be able to register with a unique email and password.
- Login functionality with role-based access control (Admin, Organizer, Attendee).
- Password validation and simple encryption for security.
- Forgot password/reset option (optional for bonus).

2. Event Management

- Admin and Organizers can create new events with details including:
 - Event Name
 - Event Description
 - Date & Time
 - Venue / Location
 - Maximum number of participants (optional)
 - Edit and delete existing events.
 - View list of all upcoming and past events.
 - Filter and search events by date, type, or organizer.

3. Registration Management

- Attendees can view event details before registration.
- Registration for events with confirmation messages.
- Maintain a list of participants per event, including their details (name, email, contact).
- Optional: Waitlist management if event reaches maximum capacity.

4. Admin Management

- Admin can add, update, and remove users.
- Assign roles to users (Organizer, Attendee).
- Generate reports such as:
 - Number of events per organizer
 - Total attendees per event
 - Daily/weekly registrations

- Monitor system usage and activities.

5. Daily Activity Tracking

- Track all events created per day.
- Track all registrations per day.
- Provide a summary of daily activities, including totals and basic statistics.

Non-Functional Requirements

1. Usability

- GUI should be simple and intuitive, using Java Swing components.
- Easy navigation between modules (Events, Users, Registrations).
- Clear error messages and user guidance for invalid actions.

2. Reliability

- The system should handle invalid inputs gracefully.
- System should be consistent and prevent data duplication.
- GUI should remain stable under normal use.

3. Performance

- GUI should respond quickly to user interactions.
- Efficient handling of in-memory lists for events, users, and registrations.

No noticeable lag while displaying event lists or registrations.

4. Maintainability

- Modular design using Java classes for each functional module.
- Proper naming conventions and documentation within code.
- Easy to update and add new features in future.

5. Portability

- The application should run on any machine with Java installed.
- No external dependencies apart from Java SDK.

6. Security

- Role-based access control: different functionalities available based on user role.
- Password protection for login.
- Basic validation to prevent unauthorized access.

7. Documentation & Reporting

- Include detailed UML diagrams: Use Case, Class, Sequence, and Activity diagrams.
- Proper in-code comments explaining the logic.
- Generate simple text-based or GUI-based summaries of events and registrations.